

A NON-REDUCED CHARACTERISTIC-ZERO ALGEBRA WITH VANISHING COTANGENT COMPLEX

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ABSTRACT. For every field k of characteristic zero, we construct a local algebraic k -algebra A with $L_{A/k} \simeq 0$, $A_{\text{red}} \cong k$, and a nonzero square-zero element. In particular, A is not ind-étale over k . The construction is a sequential filtered colimit of finite local complete intersections: the transition maps annihilate cotangent homology, while an explicit exponent bound preserves a chosen nilpotent. It answers Bhatt's existence question and the non-reduced formulation of Mondal and Mukhopadhyay affirmatively. For $k = \mathbb{Q}$, it also realizes the case $A_{\text{red}} \cong \mathbb{Q}$ singled out in Antieau's Question 6.

1. INTRODUCTION

Let k be a field of characteristic zero. Bhatt asked whether there exists a k -algebra A such that $L_{A/k} \simeq 0$ but A is not ind-étale over k [2, Question 0.3]. Mondal and Mukhopadhyay proved that a reduced k -algebra A with $\Omega_{A/k} = 0$ is ind-étale [4, Theorem 2.10]. They accordingly formulated the remaining existence question in terms of non-reduced algebras with vanishing cotangent complex [4, Question 1.4]. We construct such an algebra.

Theorem 1. *For every field k of characteristic zero there is a local, algebraic k -algebra A , with residue field k , containing a nonzero element z such that*

$$L_{A/k} \simeq 0, \quad z^2 = 0.$$

The algebra A can be taken to be a sequential filtered colimit of finite local complete intersections over k .

For the algebra constructed below, the nilradical is the maximal ideal, so $A_{\text{red}} \cong k$. Since ind-étale algebras over a field are reduced, Theorem 1 gives an affirmative answer to Bhatt's existence question. Equivalently, it disproves the implication

$$L_{A/k} \simeq 0 \implies A \text{ is ind-étale over } k.$$

Antieau's Question 6 asks for a non-reduced $\mathbb{Q}$ -algebra with vanishing cotangent complex, and specifically raises the possibility that its reduction is $\mathbb{Q}$ [1, Question 6]. Taking $k = \mathbb{Q}$ gives precisely this case. The maximal ideal of our algebra also satisfies $\mathfrak{m}_A^2 = \mathfrak{m}_A$, as follows directly from the transition maps.

Gabber constructed non-reduced local algebras in characteristic zero with vanishing Kähler differentials; see Mukhopadhyay and Smith [5, Theorem 2.2]. Their starting algebra, a finite-dimensional Jacobian quotient of a power series ring, contains a nonzero square-zero element with zero differential [5, Lemma 2.4]. The construction below also annihilates first cotangent homology while retaining a nilpotent through the transition maps.

The construction uses finite local complete intersections in which a power of the product of the variables has a prescribed nilpotency index (Lemma 2). This power annihilates cotangent homology in degrees zero and one, and every degree-one cycle has its coefficients in the maximal ideal (Lemma 3). Consequently, a map carrying the source maximal ideal into the ideal generated by the target power annihilates cotangent homology (Lemma 4). Proposition 6 constructs such maps and computes how they change the exponent of the distinguished power. The estimate (6) keeps the exponent representing the image of an initial square-zero element below the corresponding nilpotency index at every stage. This

element survives in the colimit, while cotangent homology vanishes by compatibility with filtered colimits.

Throughout, rings are commutative and unital, and k is a field of characteristic zero. We use homological indexing, so $H_0(L_{R/k}) = \Omega_{R/k}$. All cotangent complexes and differentials are relative to k . For a local ring R , its maximal ideal is denoted by $\mathfrak{m}_R$, and $R_{\text{red}} = R/\sqrt{0}$.

2. FINITE LOCAL COMPLETE INTERSECTIONS

Fix integers

$$d \geq 2, \quad a \geq 2, \quad q > ad.$$

Let

$$P = k[x_1, \dots, x_d]_{(x_1, \dots, x_d)}, \quad U = \prod_{i=1}^d x_i,$$

and define

$$r_i = x_i^{q-1} - U^a/x_i, \quad B(d, a, q) = P/(r_1, \dots, r_d).$$

Expressions such as U^a/x_i denote monomials in P , and the same symbols denote their residue classes in $B(d, a, q)$. The relations r_i are the partial derivatives of

$$\frac{1}{q} \sum_{i=1}^d x_i^q - \frac{1}{a} U^a.$$

Lemma 2. *The algebra $B(d, a, q)$ is a finite local complete intersection with residue field k . The monomial U^{ad-2} is a nonzero socle generator. In particular,*

$$U^{ad} = 0, \quad U^{a(d-1)} \neq 0.$$

Proof. Multiplying r_i by x_i gives $x_i^q = U^a$ in $B(d, a, q)$. Taking the product over i yields $U^q = U^{ad}$. Since $q - ad > 0$, the element $1 - U^{q-ad}$ is a unit, and hence

$$U^{ad} = 0, \quad x_i^{qd} = 0 \quad (1 \leq i \leq d).$$

Thus $B(d, a, q)$ is finite over k , with residue field k , and $(r_1, \dots, r_d)$ is $\mathfrak{m}_P$ -primary. The relations form a system of parameters in the Cohen–Macaulay ring P , so they form a regular sequence.

It remains to show that $U^{a(d-1)}$ is nonzero. Since $a \geq 2$, the diagonal expression

$$r_i = x_i \left(x_i^{q-2} - U^a/x_i^2 \right)$$

gives the socle determinant

$$E = \prod_{i=1}^d \left(x_i^{q-2} - U^a/x_i^2 \right).$$

By the determinant description of the socle of a zero-dimensional complete intersection, $E \neq 0$ and $\text{Soc}(B(d, a, q)) = kE$; see [3, Lemma 4].

The nonvanishing can also be read from the map

$$\text{Ext}_P^d(k, P) \longrightarrow \text{Ext}_P^d(B(d, a, q), P)$$

induced by the residue map. It is injective because $\text{Ext}_P^{d-1}(\mathfrak{m}_{B(d, a, q)}, P) = 0$. The corresponding vanishing for every finite-length P -module follows by a composition series from the Koszul resolution of k . The Koszul resolutions on (x_i) and (r_i) identify the source and target with k and $B(d, a, q)$, respectively; comparing them identifies the injection with $1 \mapsto E$. Its image is annihilated by the maximal ideal; the socle is one-dimensional because $B(d, a, q)$ is a zero-dimensional complete intersection.

We compute E in $B(d, a, q)$. The term taking the first summand in every factor is U^{q-2} . For a nonempty subset of s indices, take the second summand precisely in those factors. The unsigned term has exponent $as - 2$ on the chosen variables and exponent $q + as - 2$

on the others. As $as \geq 2$, replacing one factor x_j^q by U^a for each unchosen variable gives U^{ad-2} . Therefore

$$\begin{aligned} E &= U^{q-2} + \sum_{s=1}^d (-1)^s \binom{d}{s} U^{ad-2} \\ &= U^{q-2} - U^{ad-2} = (U^{q-ad} - 1)U^{ad-2}. \end{aligned}$$

The factor $U^{q-ad} - 1$ is a unit. Hence U^{ad-2} is also a nonzero socle generator. In particular $U^{q-ad}U^{ad-2} = 0$, and $E = -U^{ad-2}$. Finally,

$$U^{ad-2} = U^{a(d-1)}U^{a-2} \neq 0,$$

so $U^{a(d-1)} \neq 0$. □

3. COTANGENT HOMOLOGY AND TRANSITION MAPS

Localization gives $L_{P/k} \simeq \Omega_{P/k}[0]$, with $\Omega_{P/k}$ free on $dx_1, \dots, dx_d$; see [6, Tags 08R2 and 08R3]. The regular-sequence presentation therefore identifies $L_{B(d,a,q)/k}$ with

$$(1) \quad [B(d, a, q)^d \xrightarrow{\mathcal{J}} B(d, a, q)^d], \quad \mathcal{J}(\alpha)_j = \sum_{i=1}^d \alpha_i \frac{\partial r_i}{\partial x_j},$$

in degrees 1, 0, whose bases are the relation classes and the variable differentials, respectively; see [6, Tags 08RB and 08SL]. Thus $H_1(L_{B(d,a,q)/k}) = \ker \mathcal{J}$, and all higher homology groups vanish.

Lemma 3. *In $B(d, a, q)$ one has*

$$d(U^a) = 0, \quad U^a \Omega_{B(d,a,q)/k} = 0, \quad U^a H_1(L_{B(d,a,q)/k}) = 0.$$

Every vector in $\ker \mathcal{J}$ has all its coordinates in the maximal ideal.

Proof. Differentiating $x_i^q = U^a$ and using $x_i^{q-1} = U^a/x_i$ gives

$$q(U^a/x_i) dx_i = d(U^a).$$

On the other hand, $d(U^a) = a \sum_i (U^a/x_i) dx_i$. Summation therefore gives $(q - ad)d(U^a) = 0$. Since $q - ad$ and q are invertible in k , it follows that $d(U^a) = 0$ and $(U^a/x_i) dx_i = 0$ for every i . Multiplication by x_i proves $U^a \Omega_{B(d,a,q)/k} = 0$.

Let $\alpha \in \ker \mathcal{J}$. The monomial identity

$$x_j \frac{\partial(U^a/x_i)}{\partial x_j} = (a - \delta_{ij})U^a/x_i$$

gives

$$0 = x_j \mathcal{J}(\alpha)_j = q\alpha_j(U^a/x_j) - a \sum_i \alpha_i(U^a/x_i).$$

Summing over j yields

$$(q - ad) \sum_i \alpha_i(U^a/x_i) = 0.$$

Consequently $\alpha_j(U^a/x_j) = 0$ for every j . Multiplication by x_j shows that $U^a \alpha_j = 0$ coordinatewise, proving $U^a H_1(L_{B(d,a,q)/k}) = 0$.

For each j ,

$$U^{ad-2} = (U^a/x_j)x_j U^{a(d-1)-2}.$$

The last exponent is nonnegative, and the left-hand side is nonzero by Lemma 2. Thus $U^a/x_j \neq 0$. As α_j annihilates this monomial, it cannot be a unit. □

A map carrying the source maximal ideal into the ideal generated by the target power annihilates the differential of every source element. The condition on degree-one cycles in Lemma 3 also forces the map on H_1 to vanish.

Lemma 4. *Let*

$$f : B(d, a, q) \longrightarrow B(D, b, Q)$$

be a local k -algebra map between two members of the family. Write $y_1, \dots, y_D$ for the target variables and $V = \prod_{j=1}^D y_j$. If

$$f(\mathfrak{m}_{B(d,a,q)}) \subseteq V^b B(D, b, Q),$$

then the natural maps

$$H_j(L_{B(d,a,q)/k}) \longrightarrow H_j(L_{B(D,b,Q)/k})$$

are zero for every $j \geq 0$.

Proof. For $s \in B(D, b, Q)$, Lemma 3 gives

$$d(V^b s) = s d(V^b) + V^b ds = 0.$$

Thus the map on H_0 is zero. Both algebras are complete intersections, so it remains to treat H_1 .

Let P' be the target local polynomial ring, and let $J \subset P'$ be the ideal generated by its defining relations $r'_1, \dots, r'_D$. Choose lifts $F_i \in P'$ of the images of the source variables. The conormal sequence and $df(x_i) = 0$ give $b_{i\ell} \in P'$ with

$$dF_i - \sum_{\ell} b_{i\ell} dr'_{\ell} \in J\Omega_{P'/k}.$$

Replacing F_i by $F_i - \sum_{\ell} b_{i\ell} r'_{\ell}$ gives

$$d\left(F_i - \sum_{\ell} b_{i\ell} r'_{\ell}\right) = dF_i - \sum_{\ell} b_{i\ell} dr'_{\ell} - \sum_{\ell} r'_{\ell} db_{i\ell} \in J\Omega_{P'/k}.$$

The corrected lifts still lie in $\mathfrak{m}_{P'}$, and hence define a local map $\tilde{f} : P \rightarrow P'$ lifting f .

Each $\tilde{f}(r_i)$ lies in J , and the chain rule gives $d\tilde{f}(r_i) \in J\Omega_{P'/k}$. Thus

$$[\tilde{f}(r_i)] \in \ker(J/J^2 \longrightarrow \Omega_{P'/k} \otimes_{P'} B(D, b, Q)) = H_1(L_{B(D,b,Q)/k}).$$

The map of conormal complexes sends a source cycle $\alpha = (\alpha_i)$ to

$$\sum_i f(\alpha_i) [\tilde{f}(r_i)].$$

The natural comparison of the conormal model with the cotangent complex is compatible with localization. Thus this formula gives the natural homology map; see [6, Tags 00S1 and 08RB]. Changing the lifts by elements $h_j \in J$ changes the image of $[r_i]$ in J/J^2 by

$$\sum_j f\left(\frac{\partial r_i}{\partial x_j}\right) [h_j].$$

After multiplication by $f(\alpha_i)$ and summation over i , this change vanishes by the cycle equations.

By Lemma 3, each α_i lies in the source maximal ideal. Hence $f(\alpha_i) \in V^b B(D, b, Q)$. The same lemma shows that V^b annihilates $H_1(L_{B(D,b,Q)/k})$, so every term of the displayed image is zero. $\square$

Remark 5. Lemma 4 concerns the natural homology maps of k -vector spaces. It makes no assertion that the corresponding derived-base-changed morphism of cotangent complexes is zero. These homology maps are the ones used in the filtered-colimit argument in Section 5.

4. EXPLICIT TRANSITION MAPS

Fix d, a, q as in Section 2 and an integer $b \geq 2$. Set

$$(2) \quad \ell = (a-1)q + a(d-1).$$

The identity

$$(3) \quad bq(q-1) - bd\ell = b(q-ad)(q+d-1) > 0$$

ensures that $B(d\ell, b, bq(q-1))$ belongs to the family. Index its variables by y_{ij} , where $1 \leq i \leq d$ and $1 \leq j \leq \ell$, and write

$$Y_i = \prod_{j=1}^{\ell} y_{ij}, \quad V = \prod_{i=1}^d Y_i.$$

Proposition 6. *The assignments*

$$(4) \quad x_i \mapsto V^b Y_i^{b(q-1)}$$

define a local k -algebra map

$$\Phi : B(d, a, q) \longrightarrow B(d\ell, b, bq(q-1)).$$

They satisfy

$$\Phi(U^a) = V^{b(\ell+q)}, \quad \Phi(\mathfrak{m}_{B(d,a,q)}) \subseteq V^b B(d\ell, b, bq(q-1)).$$

The induced maps on cotangent homology are zero in every degree.

Proof. The relations in the target give

$$y_{ij}^{bq(q-1)-1} = V^b / y_{ij}.$$

Taking the product over the i th block yields

$$Y_i^{bq(q-1)-1} = V^{b\ell} / Y_i.$$

Multiply this equality by $Y_i \prod_{h \neq i} Y_h^{b(q-1)}$. The left-hand side becomes the image of x_i^{q-1} under (4). The right-hand side is $V^{b\ell} \prod_{h \neq i} Y_h^{b(q-1)}$. It is the image of U^a / x_i : the identity

$$\ell = ad - 1 + (a-1)(q-1)$$

shows that substituting (4) into $x_i^{a-1} \prod_{h \neq i} x_h^a$ gives

$$\begin{aligned} & Y_i^{b[ad-1+(a-1)(q-1)]} \prod_{h \neq i} Y_h^{b[ad-1+a(q-1)]} \\ &= Y_i^{b\ell} \prod_{h \neq i} Y_h^{b\ell+b(q-1)}. \end{aligned}$$

Thus each source relation vanishes in the target. The images of the source variables lie in the target maximal ideal, so the map extends to the localizations and induces Φ .

Taking products in (4) gives

$$\Phi(U) = V^{b(d+q-1)}, \quad \Phi(U^a) = V^{ba(d+q-1)} = V^{b(\ell+q)}.$$

The image of the source maximal ideal lies in the ideal generated by V^b . Lemma 4 therefore gives the asserted vanishing on cotangent homology. $\square$

5. THE COLIMIT AND THE SURVIVING NILPOTENT

Proof of Theorem 1. We choose the parameters in Proposition 6 so that, at each stage, the exponent representing the image of the initial distinguished power is smaller than its nilpotency index. Set

$$a_n = 2^{n+3} \quad (n \geq 0), \quad d_0 = 2, \quad q_0 = 17,$$

and define recursively

$$(5) \quad \begin{aligned} \ell_n &= (a_n - 1)q_n + a_n(d_n - 1), \\ d_{n+1} &= d_n \ell_n, \\ q_{n+1} &= a_{n+1}q_n(q_n - 1). \end{aligned}$$

Since $q_0 = 17 > 16 = a_0 d_0$, equation (3) proves inductively that $q_n > a_n d_n$. Applying Proposition 6 with $b = a_{n+1}$ gives maps

$$\Phi_n : B(d_n, a_n, q_n) \longrightarrow B(d_{n+1}, a_{n+1}, q_{n+1}) \quad (n \geq 0).$$

Write $x_{n,1}, \dots, x_{n,d_n}$ for the variables at stage n . In the application of Proposition 6 defining Φ_n , identify the target variable y_{ij} with $x_{n+1,(i-1)\ell_n+j}$ for $1 \leq i \leq d_n$ and $1 \leq j \leq \ell_n$. Set

$$U_n = \prod_{i=1}^{d_n} x_{n,i}, \quad A = \operatorname{colim}_n B(d_n, a_n, q_n).$$

A nonzero square-zero element. Lemma 2 gives $U_n^{a_n d_n} = 0$ and $U_n^{a_n(d_n-1)} \neq 0$. In particular, $U_0^{a_0}$ is nonzero and square-zero. Define positive integers

$$e_0 = 1, \quad e_{n+1} = (\ell_n + q_n)e_n.$$

Proposition 6 shows that its image at stage n is $U_n^{a_n e_n}$. The recurrence gives

$$\frac{e_{n+1}}{d_{n+1}} = \frac{e_n}{d_n} \left(1 + \frac{q_n}{\ell_n}\right) \leq \frac{e_n}{d_n} \frac{a_n}{a_n - 1}.$$

For nonnegative u_j with sum at most one, induction gives $\prod_j (1 - u_j) \geq 1 - \sum_j u_j$ for every finite product. Since $\sum_{n \geq 0} a_n^{-1} = 1/4$, it follows that

$$(6) \quad \frac{e_n}{d_n} \leq \frac{1}{2} \prod_{j < n} \left(1 - \frac{1}{a_j}\right)^{-1} \leq \frac{1}{2} \left(1 - \frac{1}{4}\right)^{-1} = \frac{2}{3} < 1.$$

Hence $1 \leq e_n \leq d_n - 1$. If $U_n^{a_n e_n}$ were zero, multiplication by $U_n^{a_n(d_n-1-e_n)}$ would contradict Lemma 2. Thus the image of $U_0^{a_0}$ is nonzero at every stage. An element represented at one stage of a filtered colimit is zero precisely when its image is zero at a later stage. Its image $z \in A$ is therefore nonzero, and $z^2 = 0$.

Vanishing of the cotangent complex. Proposition 6 gives zero maps

$$H_j(L_{B(d_n, a_n, q_n)/k}) \longrightarrow H_j(L_{B(d_{n+1}, a_{n+1}, q_{n+1})/k}) \quad (j \geq 0).$$

Using the functorial standard simplicial polynomial resolutions, formation of the cotangent complex commutes with filtered colimits [6, Tag 08S9]. On the underlying complexes of k -vector spaces this gives

$$\operatorname{colim}_n L_{B(d_n, a_n, q_n)/k} \simeq L_{A/k}.$$

Filtered colimits of vector spaces are exact. Consequently,

$$H_j(L_{A/k}) \cong \operatorname{colim}_n H_j(L_{B(d_n, a_n, q_n)/k}) = 0 \quad (j \geq 0).$$

Thus $L_{A/k} \simeq 0$.

The local algebra and its reduction. The residue maps $B(d_n, a_n, q_n) \rightarrow k$ are compatible with Φ_n , so they induce an augmentation $A \rightarrow k$. Every element of A can be written as $c + \nu$, where $c \in k$ and ν is nilpotent, by representing it at a finite stage. If $c \neq 0$, a finite geometric series gives an inverse for $c + \nu$; if $c = 0$, the element is nilpotent. Hence the kernel of the augmentation is both the nilradical and the unique maximal ideal. In particular, the residue field and A_{red} are both k . Moreover, $c + \nu$ satisfies $(X - c)^N = 0$ for some N , so A is algebraic over k . This proves the theorem.

Each variable image in (4) belongs to the square of the target maximal ideal. Thus every element of $\mathfrak{m}_A$, represented at a finite stage, lies in $\mathfrak{m}_A^2$ after passage to the next stage. It follows that $\mathfrak{m}_A = \mathfrak{m}_A^2$. $\square$

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